Answers learning task 7

2026 – 2027

Exercise 5.1: Show that the following function satisfies the properties of a joint probability mass function.

x	y	$f_{XY}(x,y)$
1	1	$1/4$
1.5	2	$1/8$
1.5	3	$1/4$
2.5	4	$1/4$
3	5	$1/8$

Determine the following:

(a) $P(X < 2.5, Y < 3)$
(b) $P(X < 2.5)$
(c) $P(Y < 3)$
(d) $P(X > 1.8, Y > 4.7)$
(e) $E(X)$, $E(Y)$, $V(X)$, $V(Y)$

Answer

(a) 0.375
(b) 0.625
(c) 0.375
(d) 0.125
(e) $E(X) = 1.8125, \quad E(Y) = 2.875, \quad V(X) = 0.4961, \quad V(Y) = 1.8594$

Exercise 5.5: In the transmission of digital information, the probability that a bit has high, moderate and low distortion is 0.01, 0.04, and 0.95, respectively. Suppose that three bits are transmitted and that the amount of distortion of each bit is assumed to be independent. Let X and Y denote the number of bits with high and moderate distortion out of the three, respectively. Determine:

(a) $f_{XY}(x,y)$
(b) $f_X(x)$
(c) $E(Y)$
(d) $f_{Y|2}(y)$
(e) $E(Y|X = 1)$
(f) Are X and Y independent?

Answer

(a)
$(0,0) : 0.857375, \quad (0,1) : 0.1083, \quad (0,2) : 0.00456, \quad (0,3) : 0.000064,$
$(1,0) : 0.027075, \quad (1,1) : 0.00228, \quad (1,2) : 0.000048, \quad (1,3) : 0,$
$(2,0) : 0.000285, \quad (2,1) : 0.000012, \quad (2,2) : 0, \quad (2,3) : 0,$
$(3,0) : 0.000001, \quad (3,1) : 0, \quad (3,2) : 0, \quad (3,3) : 0$

(b) $f_X(0) = 0.970299$, $f_X(1) = 0.029403$, $f_X(2) = 0.000297$, and $f_X(3) = 0.000001$.

(c) 0.12

(d) $f_{Y|2}(0) = 0.9596$ and $f_{Y|2}(1) = 0.0404$

(e) 0.0808

(f) No

Exercise 5.31: Determine the value for c and the covariance and correlation for the joint probability mass function $f_{XY}(x, y) = c(x + y)$ for $x = 1, 2, 3$ and $y = 1, 2$.

Answer

$c = \frac{1}{21}$, $\text{cov}(X, Y) = -0.0136$, $\rho(X, Y) = -0.0346$

Exercise 5.39: The joint probability distribution is

x	-1	0	0	1
y	0	-1	1	0
$f_{XY}(x, y)$	$1/6$	$1/3$	$1/3$	$1/6$

Answer

$\text{cov}(X, Y) = 0$, X, Y dependent, because $P(X = 0) \cdot P(Y = 0) = \frac{2}{9} \neq P(X = 0, Y = 0) = 0$

Exercise 5.85: Show that the following function satisfies the properties of a joint probability mass function:

x	y	$f(x, y)$
0	0	$1/4$
0	1	$1/8$
1	0	$1/8$
1	1	$1/4$
2	2	$1/4$

Determine the following:

(a) $P(X < 0.5, Y < 1)$
(b) $P(X \leq 1)$
(c) $P(X < 1.5)$
(d) $P(X > 0.5, Y < 1.5)$
(e) $E(X), E(Y), V(X), V(Y)$
(f) Marginal probability distribution of the random variable X
(g) Conditional probability distribution of Y given that $X = 1$
(h) $E(Y|X = 1)$
(i) Are X and Y independent? Why or why not?
(j) Correlation between X and Y

Answer

(a) 0.25
(b) 0.75
(c) 0.75
(d) 0.375
(e) $E(X) = 0.875, \quad E(Y) = 0.875, \quad V(X) = 0.6094, \quad V(Y) = 0.6094$
(f) $f_X(0) = 0.375, \quad f_X(1) = 0.375, \quad f_X(2) = 0.25$
(g) $f_{Y|1}(0) = 1/3, \quad f_{Y|1}(1) = 2/3$
(h) $2/3$
(i) No, because $f_{XY}(1,1) = 1/4 \neq f_X(1)f_Y(1) = 9/64$
(j) 0.7949